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REVIEWERS: ARMY - MO
AIR FORCE - 85
--69

USERS: NAVY- YD
- MC

DD FORM 672-1 (Superseding)

P. A. ARMY - MO(ERDL)
Other Cust.
NAVY - YD
AIR FORCE - 69

PROCUREMENT SPECIFICATION
MIL-S-3785

SUPERSEDES:

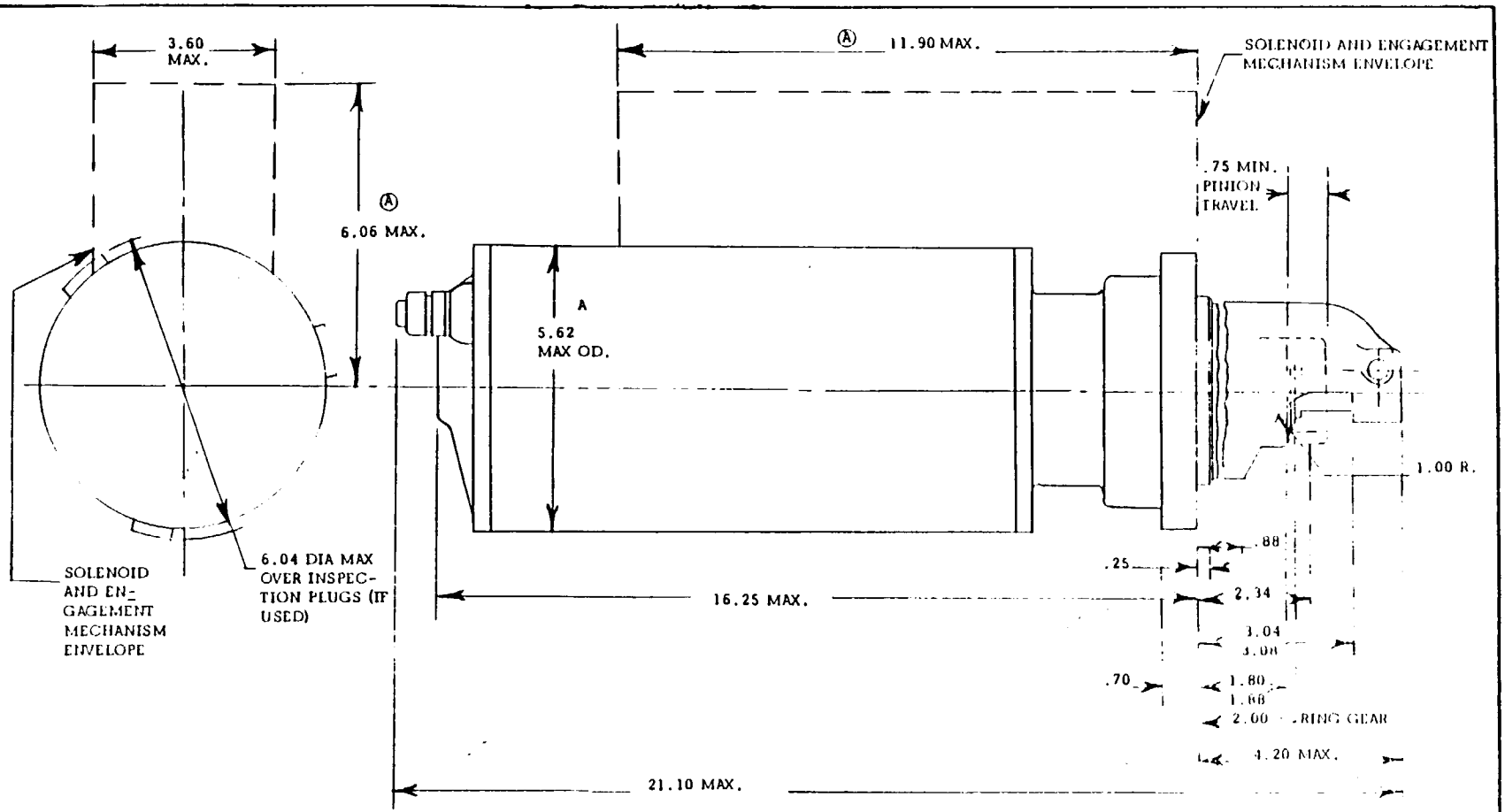
TITLE
STARTER, ENGINE, ELECTRICAL, 24-VOLT, D.C., 5-9/16 - INCH FRAME DIAMETER, SOLENOID-ACTUATED, HEAVY-DUTY

MILITARY STANDARD

MS 53008

SHEET 1 OF 4

FED. SUP CLASS 2920



NOTES:

1. PINION DATA:
   11 TEETH; 6/8 PITCH (12 TOOTH BLANK); 2.000 P.D.; 2.250 O.D.; 20° PRESSURE ANGLE; .015 TO .025 P.L. CLEARANCE.
2. MOTOR IS INSULATED TYPE MOTOR.
3. MOUNTING IS SIMILAR TO #3 SAE FLANGE MOUNTINGS.
4. Ⓐ TERMINAL THREAD SIZE AND LENGTH:
   BATTERY - 1/2 - 13 UNC 2A, MINIMUM LENGTH - 1.50
   MOTOR GROUND - 1/2 - 13 UNC 2A, MINIMUM LENGTH - 1.00
   SWITCH - #10 - 32 UNF 2A
5. DESIGN AND ARRANGEMENT OF SOLENOID AND ENGAGEMENT MECHANISM ASSEMBLY ARE OPTIONAL, PROVIDED SHIFT LEVER AND LINKAGE ARE INCLOSED AND ALSO PROVIDED TERMINAL THREAD SIZES CONFORM TO AND LIMITING OVERALL DIMENSIONS DO NOT EXCEED THOSE SPECIFIED HEREIN
6. Ⓐ ROTATION VIEWING DRIVE END - CLOCKWISE.
7. Ⓐ TOLERANCE: DIMENSIONS: ± .030 UNLESS OTHERWISE SPECIFIED.
   ANGULAR: ± 1°.
8. THIS MOTOR IS IDENTIFIED AS TYPE II (RING GEAR ENGAGEMENT), GRADE A (WATERPROOF), CLASS 1 (POSITIVE DRIVE) OF REFERENCED SPECIFICATION. Ⓐ

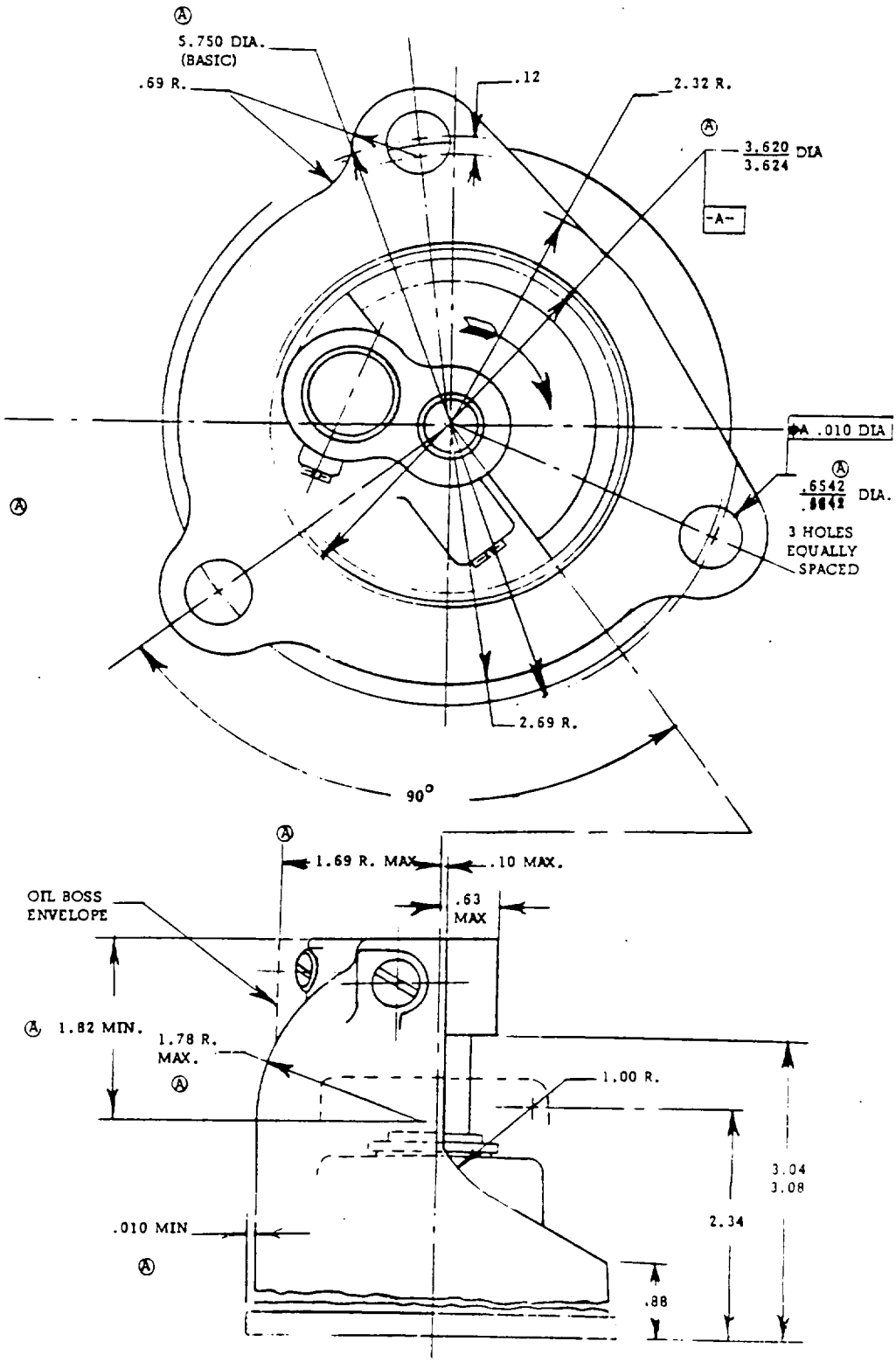
(NOTES CONTINUED ON SHEET 4)

Ⓐ FOR CHANGES SEE SHEETS 1 THRU 4

APPROVED 2 May 1962 REVISED Ⓐ 28 January 1965

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APPROVED 2 May 1962

REVIEWERS: ARMY - MO
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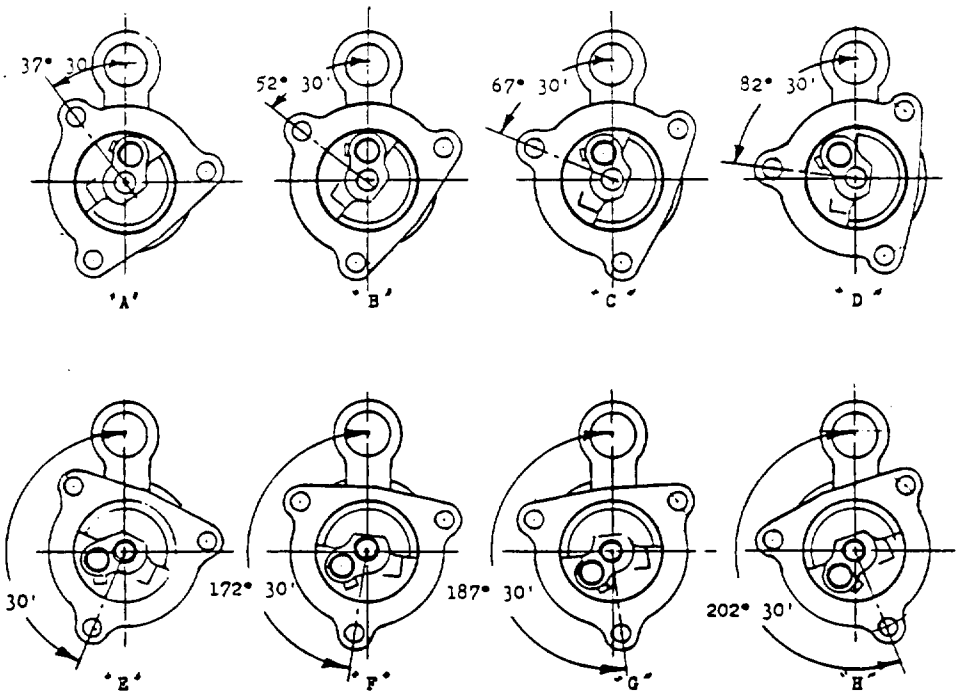
P.A. ARMY - MO(ERDL) Other Cust. NAVY - YD AIR FORCE - 69	TITLE STARTER, ENGINE, ELECTRICAL, 24-VOLT D.C., 5-9/16-INCH FRAME DIAMETER, SOLENOID-ACTUATED, HEAVY-DUTY	MILITARY STANDARD MS 53008
PROCUREMENT SPECIFICATION MIL-S-3785	SUPERSEDES.	SHEET 2

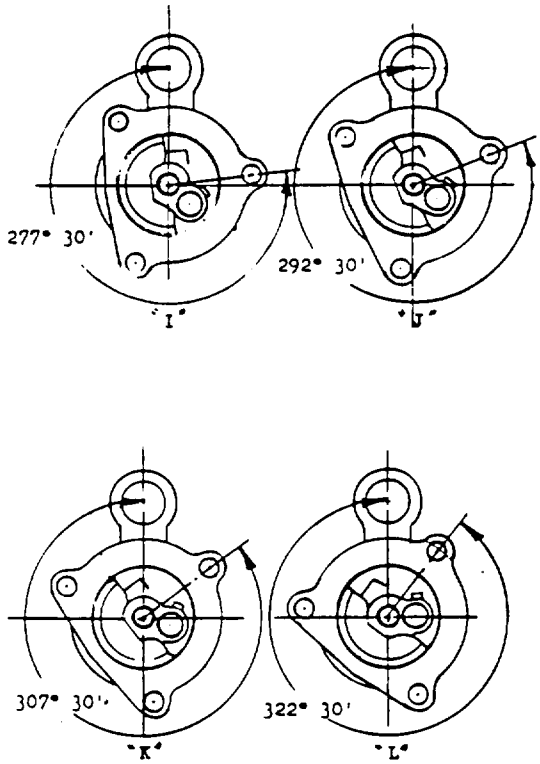
DD FORM 1 SEP 57 672-1 (Coordinated)

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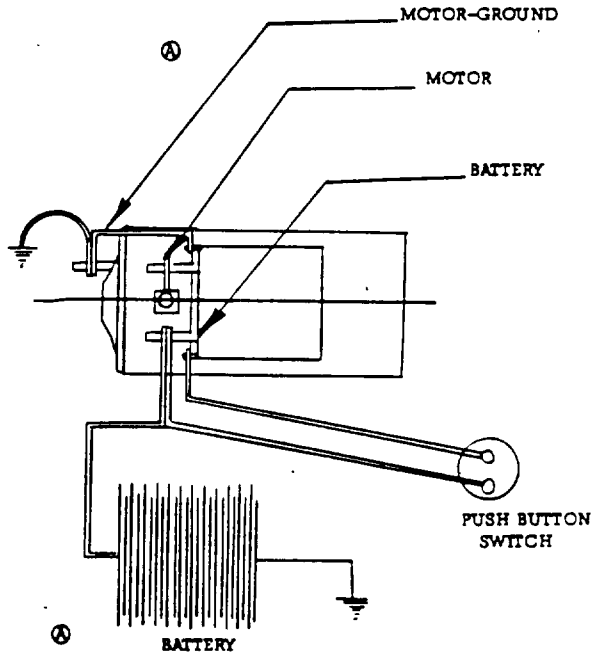
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MOUNTING FLANGE AND PINION HOUSING SHALL BE ADJUSTABLE TO ALL POSITIONS SHOWN HEREON. MOUNTING POSITIONS VIEWING DRIVE END OF MOTOR.



TYPICAL WIRING DIAGRAM

REVIEWERS: ARMY - MC
AIR FORCE - 85
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USERS: NAVY - YD
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APPROVED 2 May 1962 REVISED Ⓐ FOR CHANGES SEE SHEETS 1 THRU 4

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PROCUREMENT SPECIFICATION MIL-S-3785	SUPERSEDES:	SHEET 3

DD FORM SEP 67 672-1 (Coordinated)

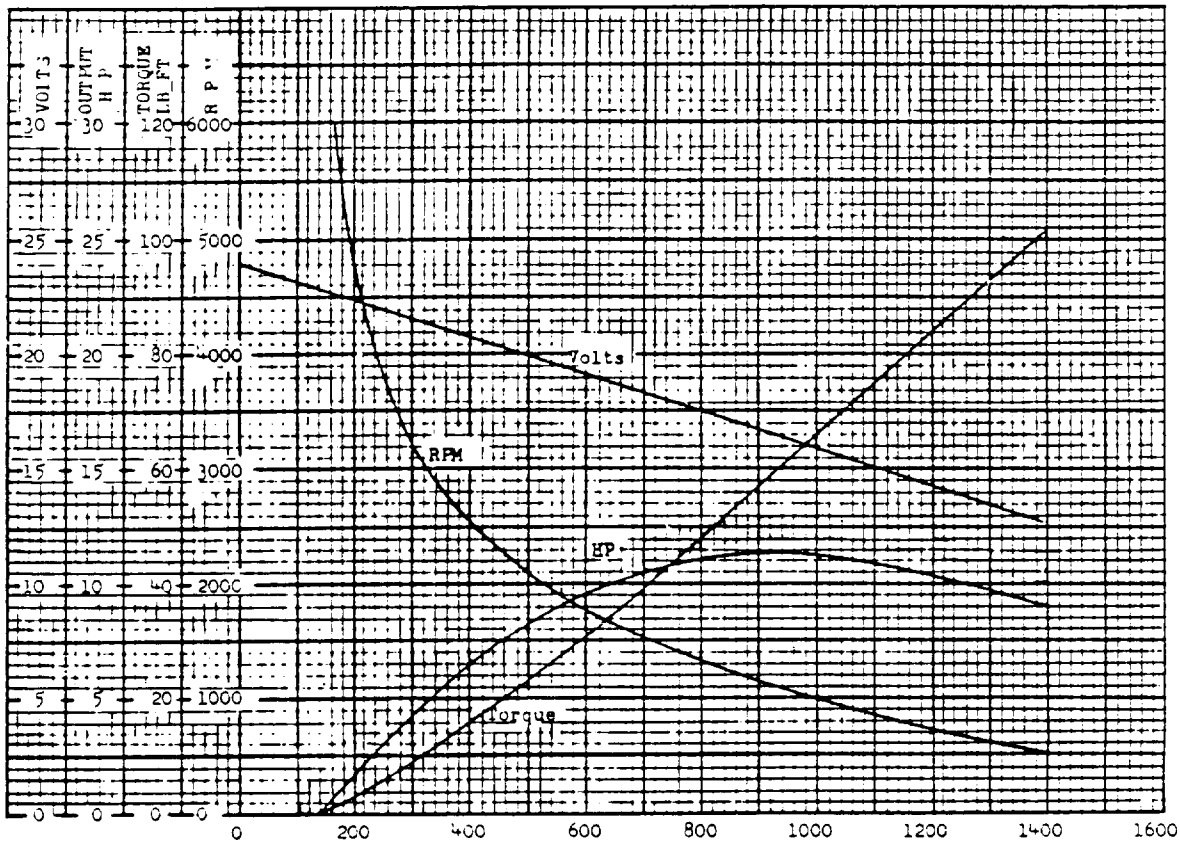
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REVIEWERS: ARMY - MO
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- 69
USERS: NAVY - YD
- MC



Ⓐ PERFORMANCE CURVE

(CONTINUATION OF NOTES FROM SHEET 1)

9. MOTOR SHALL BE EQUIPPED WITH OIL SEAL FOR WET CLUTCH OPERATION.
10. UNLESS OTHERWISE SPECIFIED, ALL DIMENSIONS ARE IN INCHES.
11. MODIFICATIONS IN OUTSIDE APPEARANCE AND CONSTRUCTION ARE ACCEPTABLE IF UNIT INTERCHANGEABILITY IS NOT AFFECTED.
(A 12. FOR DESIGN FEATURE PURPOSES THIS STANDARD TAKES PRECEDENCE OVER PROCUREMENT DOCUMENTS REFERENCED HEREIN.
A 13. REFERENCED DOCUMENTS SHALL BE OF THE ISSUE IN EFFECT ON DATE OF INVITATIONS FOR BID.
14. TORQUE AND R.P.M. MUST NOT BE LESS THAN 10% BELOW THAT SHOWN BY CURVES, WITH VOLTAGE APPLIED AS SPECIFIED. OUTPUT CURVE IS NOT RELATED TO TORQUE AND R.P.M. CURVES SHOWN, BUT IS THE MINIMUM FOR ANY PARTICULAR MOTOR.
15. THE VOLTAGE CURVE SHOWN IS THE SAE HEAVY DUTY 24-VOLT BATTERY CURVE.
A 16. LOCATION OF NAME PLATE SHALL BE OPTIONAL. NAMEPLATE SHALL BE FASTENED TO STARTER FRAME WITH RIVETS OR SCREWS AND LOCK WASHERS.
A 17. STALLED TORQUE DATA: VOLTS - 4.3 (APPROX)
CURRENT (AMPS) (MAX) - 900
TORQUE (LB-FT) (MIN) - 52
(A 18. JUMPER BETWEEN SOLENOID-GROUND TERMINAL AND MOTOR-GROUND TERMINAL SHALL BE FURNISHED WITH THIS STARTER.
(A 19. SOLENOID CURRENT CONSUMPTION (PULL-IN PLUS HOLD-IN) SHALL NOT EXCEED 81 AMPERES.

APPROVED 2 May 1962 REVISED Ⓐ FOR CHANGES SEE SHEETS 1 THRU

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PROCUREMENT SPECIFICATION MIL-S-1785	SUPERSEDES	SHEET 4

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